



LAB 04

I. System Requirements Context

The Equipment Borrowing Management System supports the process of requesting, approving, and tracking teaching equipment used by lecturers and students in a university. Currently, equipment is borrowed using paper forms or informal messages. This results in lost requests, double bookings, and a lack of usage reports. The system should digitalize the workflow as follows:

1: User Roles

- Borrower (Lecturer/Student):** searches for equipment, checks availability, submits borrow requests, and returns items.
- Approver (Department Head / Lab Technician):** reviews and approves borrow requests.
- Inventory Staff:** maintains equipment records, updates return status, and handles maintenance reports.
- Admin:** manages user accounts, system data, and reports.

2: Functional Requirements

- Users must **log in** to the system before submitting or approving requests.
- Borrowers can **search for equipment** and view real-time availability.
- Borrowers can **submit borrowing requests** with purpose, date/time, and quantity.
- Approvers can **approve or reject** borrow requests.
- Borrowers and staff can **view request status** (Pending / Approved / Rejected / Returned).
- The system sends **reminders for due dates** automatically.
- Inventory Staff can **record returns** and **update equipment availability**.
- Admin can **generate reports** on borrowing frequency, late returns, and utilization.

3: Non-Functional Requirements

- System must respond within **3 seconds** per transaction.
- Access is **web-based**, available to all users within the campus network.
- Data must be consistent and validated before updates.

4: Learning Objectives

- Identify and define classes, attributes, and methods from system requirements.
- Model relationships between classes (association, aggregation, composition, inheritance).
- Apply UML notation and class modeling principles.
- Verify the consistency between class diagram and system requirements.

e. Present a well-structured UML Class Diagram using standard modeling tools.

II. UML Class Diagram

Tasks

1. User (Superclass)

Attributes:

- userId: String
- name: String
- email: String
- password: String
- role: String

Methods:

- login()
- logout()
- updateProfile()
- getUserDetails()

2. Borrower (inherits User)

Methods:

- searchEquipment()
- createBorrowRequest()
- viewBorrowRequests()
- returnItems()

3. Approver (inherits User)

Methods:

- viewPendingRequests()
- approveRequest()
- rejectRequest()

4. InventoryStaff (inherits User)

Methods:

- updateEquipmentStatus()
- recordReturn()
- manageMaintenance()

5. Admin (inherits User)

Methods:

- createUser()
- updateUser()
- deleteUser()
- generateReport()

6. Equipment

Attributes:

- equipmentId: String
- name: String
- description: String
- type: String
- quantity: Integer
- condition: String

Methods:

- addEquipment()
- updateEquipment()
- getEquipmentDetails()

7. BorrowRequest

Attributes:

- requestId: String
- requestDate: Date
- purpose: String
- status: String
- borrowDate: Date
- returnDate: Date

Associations:

- borrower: Borrower
- approver: Approver
- inventoryStaff: InventoryStaff
- borrowItems: List<BorrowItem> (Composition)

Methods:

- submitRequest()
- approve()
- reject()

- `updateStatus()`
- `addBorrowItem()`

8. BorrowItem (Composite part of BorrowRequest)

Attributes:

- `itemId: String`
- `quantity: Integer`

Associations:

- `equipment: Equipment`

Methods:

- `updateQuantity()`
- `getItemInfo()`

9. Notification

Attributes:

- `notificationId: String`
- `message: String`
- `dateTime: DateTime`
- `status: String`

Associations:

- `user: User`

Methods:

- `send()`
- `updateStatus()`

10. Report

Attributes:

- `reportId: String`
- `title: String`
- `date: Date`
- `content: String`

Associations:

- `generatedBy: Admin`

Methods:

- `generate()`
- `exportPDF()`

- viewReport()

Notes on Inheritance & Associations

1. Inheritance

- User → Borrower
- User → Approver
- User → InventoryStaff
- User → Admin

→ Thể hiện phân quyền và vai trò trong EBMS.

2. Composition

- BorrowRequest ◆→ BorrowItem
→ BorrowItem không thể tồn tại nếu BorrowRequest bị xóa.

3. Aggregation / Associations

- Borrower → BorrowRequest (1..*)
- Approver → BorrowRequest (1..*)
- InventoryStaff → BorrowRequest (1..*)
- BorrowItem → Equipment (1..1)
- Notification → User (1..1)
- Report → Admin (1..1)



